

**A multi-analytic study of Reward Anticipation and Psychological Arousal as
Predictors of Productivity.**

AP Research

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Background

Productivity is most often associated with success, it is something that is highly sought after. In psychology, productivity refers to the ability of an individual to manage their time and work efficiently to maximize outputs (Psychology Today Staff, 2019). In academics, productivity plays an important role in the success of students in their classes. The overall productivity of students can be measured by assessing their academic performance (Lastiri, 2022). As a result, generally students who are more productive tend to achieve greater academic success. Despite this, many students continue to struggle with time management, which can result in low productivity which affects their learning, academic performance, and stress levels (Dhakal, 2023). As a result, learning to manage time efficiently can lead to increased productivity which can increase the academic performance of students in their classes. However, it is deeper than this, many times low productivity can be blamed on procrastination, over-commitment, poor planning skills, and distractions (Dhakal, 2023). In reality, psychological factors play a crucial role on a student's productivity levels.

Dopamine is a neurotransmitter which is released in the hypothalamus and plays a crucial role in cognitive functions and in individuals' reward system (Cristol, 2024). For this reason, high dopamine levels have been associated with increased executive functioning and working memory, which are essential for academic learning. Meanwhile, low dopamine levels are associated with low motivation and a difficulty maintaining attention, which can interfere with academic performance and impede productivity (Kamel, n.d.). In addition to the crucial role of reward anticipation through dopamine secretion, psychological arousal also plays an important

part in influencing productivity. In simple terms, psychological arousal is the state of being in relation to a stimulus, such as stress and fear (Anderson, 2024). However, arousal is not measured as a single measurement, but as a spectrum of varied levels, each of which can have significant effects on productivity. These include a low level consisting of drowsiness and a high level of increased anxiety (Psychology Town, 2025). These extremes can drastically impede productivity, for this reason optimal levels are essential for being aroused enough to be able to work effectively.

Literature Review

Many neurological and psychological factors play a big role in the regulation of many mental processes (Kamel NA, n.d.). According to the Journal of University of Anbar for Pure Science, individuals with higher dopamine levels have been identified to have better high-level cognitive processes and memory, which play a crucial role in learning and school performance. In contrast, low dopamine levels are associated with low motivation and certain difficulty in maintaining “cognitive efforts” which drastically affect attention span and motivation to learn (Kamel NA, n.d.). The relation to motivation and increased cognitive functions as a result of dopamine secretion fundamentally aligns with the human nature of participating in activities or behaviors which release significant amounts of dopamine, making that activity feel good and creating a need to repeat such activity (Cleveland Clinic, 2022). Suggesting that most of the time people are often drawn to behaviors with external rewards that makes the process desirable. This can be seen in students, who often study in order to pass an exam or achieve a positive outcome.

Psychologists have determined that behaviors are motivated by outside incentives or rewards, this is known as the Incentive Motivation Theory (Cherry, 2023). This theory was tested by Carl J. Warden from Columbia University. In the experiment a rat was placed in the start box of a short runway, and a reward was placed at the end of the runway. In order to get to the end the rat had crossed an electric grid after being food deprived. This study found that the longer the food deprived period was the more times the rat moved across (Hsiao, 2024). Highlighting the influence that external rewards have on individuals to influence extrinsic motivation and produce an action. However, while both extrinsic and intrinsic motivation have been shown to play a crucial role in the motivation to learn, research has shown that intrinsic motivation in students results in higher levels of academic achievement (James, 2025). Conveying that internal rewards such as enjoyment or curiosity have a greater impact on the academic performance of students, mainly because students are able to have a more meaningful engagement with the subject.

The Achievement Goal Theory supports the effect of internal rewards. It states that an individual's behavior is driven by their definition of success. For this reason, past research determined that students who are self motivated or intrinsically motivated are more likely to do "mastery behaviors" (Mowreader, 2025). As a result, these results from research highlight that extrinsic motivation, while having a positive effect, is less meaningful than intrinsic motivation since it ties academic performance to a physical reward creating a dependency on that reward. Once that reward is gone, the motivation to continue to learn decreases leading to a decrease in productivity (James, 2025). For this reason, it is important to recognize the benefits of both types of motivation and achieve a balance, extensive motivation helps at the start by providing the motivation to be productive, working for a short period of time. While intrinsic motivation,

provides the constant motivation to continue the task in the long term, keeping optimal productivity (Science L.E.A.F, 2024).

While reward anticipation through dopamine secretion provides a framework to influence motivation and incentivise productivity, arousal plays a major role in determining productivity levels. Peak Performance is reached when arousal levels are “sufficient” to provide “optimal” focus, motivation, and enhanced productivity (Fabritius and Hagemann, 2017). These could include the stress of completing a project or the rush of anxiety before presenting to a group of people, however when arousal levels are not at their “sweet spot” it can prove detrimental to achieving peak performance and productivity heeded (Fabritius and Hagemann, 2017). The Arousal Theory proposes that an individual's psychological arousal levels impact their performance and motivation. As such, people that want to be top performers should try to maintain an “optimal” level of arousal or stimulation (Julie, 2023).

In a study conducted in 1908, psychologists Robert Yerkes and John Dillingham Dodson discovered that when rats were subjected to mild electric stimulation, the rats ability to navigate a maze increased. However, when the stimulation increased beyond a certain threshold their ability to navigate the maze decreased quickly. The overstimulation resulted in panicked and overreacted attempts to escape from the maze, instead of the focus and alertness that they had displayed when they had been stimulated at an “optimal” level (Fabritius and Hagemann, 2017). The results of this experiment resulted in the Yerkes-Dodson law is a psychological “principle” that shows that performance increases with mental arousal, up to a point. When arousal levels are low, productivity is hindered, most commonly seen in the form of drowsiness. In contrast, when

levels are too high, most commonly in the form of high anxiety levels, productivity is hindered leading to excessive errors (Nickerson, 2023). However when the levels are moderate, arousal provides the stimulation required for focus and attention that are essential for maintaining productivity. Although this visualization is helpful for visualizing peak performance, many people respond differently to arousal levels and as a result individuals should try to see where in the spectrum they perform the best. For this reason, arousal regulation is important because it can improve productivity and achieve peak performance, as individuals are able control their “emotional and physical” states to improve productivity and create a balance (Anderson, 2024).

There is significant research on reward anticipation through dopamine secretion and psychological arousal in relation to motivation and productivity. These theories explain the relationship of the effect of dopamine secretion to influence motivation, mainly in the form of intrinsic and extrinsic motivation. These different types of motivation have been studied and determined each serves a different purpose to achieve optimal productivity and academic success. Likewise, the Motivation Theory in regards to arousal has had extensive research throughout decades, specifically in studies such as the Yerkes-Dodson experiment. It has been found and documented that arousal plays a critical role in determining productivity levels (Fabritius and Hagemann, 2017). However, there exists a significant gap when it comes to the study of reward anticipation and psychological arousal as predictors of productivity. Moreover there is very limited research on how these constructs interact in everyday academic life. It is also worth noting that most of this research excludes high school students, mainly resorting to college, employees, and testing subjects such as rats. Additionally, these findings do not accurately determine which of the two constructs determine more towards optimal productivity

levels. This is significant because without differentiation it can be assumed that both constructs contribute equally when this is not true, which can drastically affect conclusions in regards to which construct best relates to productivity. As a result, this gap highlights the need to examine the relationship between reward anticipation and psychological arousal as predictors of productivity within a high school population. This centers the purpose of this research to determine to what extent do self-reported reward anticipation and psychological alertness correlate to productivity among high school students at a high school in Southern California.

Method

In order to conduct a multi-analytic study the chosen research design consisted of a non-experimental observational study. This sampling method was chosen because it allowed for the observation of the raw variables without any manipulation, but as they appeared naturally, leading to higher external validity. Additionally, this selection reduced the possibility of any researcher bias, as no manipulation of the variables is required from the researcher. In addition, a cross-sectional survey method was used during the data collection process. This method worked well with the observational study as it analyzes data at a single point in time, providing a picture of relationship among the variables. The combination of these two methods provided a strong foundation to fit the research question because they establish a way to determine the association between psychological arousal and reward anticipation in relation to perceived productivity.

Subjects

The target population for this study consisted of high school students, especially since this population is vastly underrepresented in accordance with the literature review. The study was conducted on the responses from a high school in Southern California. The overall sample size consisted of about 251 responses. There were no requirements to participate in this study besides being a school student. For this reason, the sample was designed to be as diverse as possible, in order to properly represent the total population. The responses came from students who consented to taking the survey which was distributed by their teachers, which were emailed by the researcher regarding their availability to distribute the survey. To avoid any potential bias from teachers, a QR code was provided with instructions for students to read before starting the survey, specifically that the survey was purely voluntary.

Measurements

In order to measure the three variables (Reward Anticipation, Psychological Arousal, and Perceived Productivity), the study utilized three independent instruments, which consisted of: The BAS Reward Responsiveness Subscale for the measuring of reward anticipation, Self Assessment Manikin to assess arousal levels, and a Self-rated Performance Scale.

The BAS Reward Responsive Subscale was developed by Cravier and White in 1994 (Taubitz, 2015). This instrument was part of the larger BIS/BAS scales which was used to measure the three domains of behavioral activation (Ganesh, 2018). This study used the BAS Reward Responsive Subscale to determine the sensitivity of an individual to reward anticipation. This subscale consists of a four point scale that ranges from 1 - **very true for me** and 4 - **very false for me** (University of Miami, n.d.). The survey consisted of 5 questions that utilized the

four point scale. The individual responses were added to determine a composite score for the overall sensitivity to reward anticipation. The use of this subscale is accurate for the measurement of reward anticipation because it has proven to be reliable, showing a good internal consistency (looti, 2025) . Making it a good instrument to determine strong results.

The Self Assessment Manikin instrument uses pictures depicting various alertness levels to measure arousal in relation to an individual's reaction to a stimulus (Bradley, 1994). It consists of a 9 point scale varying from 1 - **relaxed/low arousal** to 9 **energetic/very aroused**. The survey consisted of 5 questions that utilized the nine point scale. The individual responses were added to determine a composite score for the overall alertness level for each scenario described in the question. This instrument proved highly useful to this study as it provided a way to accurately measure psychological states through descriptive images which reduced any bias and confusion that could've come from the wording describing the psychological states. The instrument has shown high reliability and validity with a lot of support for its use.

Finally, a Self-Rated Scale was used to measure the perceived productivity of individuals. This instrument was used to ensure that based on the other two instruments, individuals were able to freely determine their own perceived productivity. This was a self-developed 5 point scale, ranging from 1- **strongly disagree** to 5 - **strongly agree**. The survey consisted of 5 questions that utilized the five point scale. The individual responses were added to determine a composite score for the overall productivity. The use of this instrument was determined to be the most suitable way to measure the perceived productivity of individuals, working in accordance with the method.

Procedure

The survey used to collect the responses consisted of an informed consent, background information, and three sections relating to reward anticipation, arousal, and perceived performance, all which took about 5 minutes to complete. The survey complied with school and district policies, ensuring that students are protected from exposure to sensitive topics. Furthermore, the survey went through the process of approval by the Institutional Review Board, making sure that the survey is safe enough to not require parental consent. In addition, the survey was purely voluntary, students were reminded of this and there were optional opt outs at any time during the survey. When responding to the survey, students were presented with the reward anticipation section, then the arousal section, and finally the perceived productivity sections. This was done to maximize the accuracy of the responses so that the responses to the productivity sections reflect the same thoughts and feelings from the other two sections. To achieve the best results students were asked to make honest responses ensuring that their responses accurately reflect their feelings and thoughts. At the end of the survey participants were thanked for their responses with a thank you message at the end. Additionally, they were provided with information of how their responses would be used for the research and where they could access the research.

Ethical Considerations

In order to meet required expectations regarding the ethics of this research, specific procedures were followed to ensure that the participants's rights and beliefs are not violated.. Before participants took the survey, they were required to agree to the terms and conditions that

informed them what was involved with participating in the research. This ensured that participants are aware of any implications they might agree or disagree with. Participants were also given the opportunity to read an informed consent that guaranteed they would be taking the survey voluntarily. Students were also informed about the purpose of this study; to determine a correlation between reward anticipation and psychological arousal as predictors of productivity. Participants were not asked to provide any intimate information or information that went against school policy. However, material was provided relating to managing stress and insecurity due to academic performance, to ensure that participants had resources if for any involuntary reason the survey triggered unwanted feelings; which it should not do. Furthermore, no identifying information was collected and the responses were not shared with any individuals or third parties.

Results Overview

The survey was distributed in a period of about 3 weeks, with the final dataset consisting of a total of 241 responses. The dataset includes information regarding the participants background information as discussed in the method outline, and the responses to the 15 questions regarding Reward Anticipation, Psychological Arousal, and Perceived Productivity. The analysis of the dataset was divided into a Statistical Analysis and Computational Analysis. The Statistical Analysis focused the analysis of central tendency, internal consistency, precision and uncertainty, density, and correlation. While the Computational Analysis focused on predictive modeling using Machine Learning Models which included Random Forest and Linear Regression trained with gradient descent. The overall performance of the models was evaluated using R^2 , Mean

Absolute Error (MAE), and Root Mean Squared Error (RMSE). Additionally feature importance was also extracted from the model evaluation to identify the strongest predictors of productivity.

Reliability Measurements

In order to measure the internal consistency or reliability of the full scale and the p-values, Cronbach's alpha was measured.

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum_{i=1}^k V_i}{V_t} \right)$$

Fig 1.1 Cronbach's Alpha Formula

The formula shown in Fig 1.1, was used to determine the constants. The constant for the full dataset was determined to be $\alpha = 0.712$ which highlights an acceptable internal consistency reliability of the dataset, indicating a moderate internal-item relationship varied across the three scales. For each variable, the psychological arousal scale came the highest with an acceptable internal consistency reliability of $\alpha = 0.714$, meanwhile the perceived productivity scale showed a questionable internal consistency reliability of $\alpha = 0.653$. Conversely, the reward anticipation scale showed an unacceptable internal consistency reliability of $\alpha = 0.371$, highlighting the ineffectiveness of the items in the reward anticipation scale to measure related to a single construct. The varied consistencies found throughout the scales highlight that the scales are not measuring one single construct, but distinct and somewhat unrelated constructs. As a result, the three different constructs will be measured instead of one underlying construct because using the

overall score may hide the underperformance of the reward anticipation scale. These constructs are reward anticipation items, which measure motivation experienced in relation to receiving a reward, psychological arousal, measuring psychological alertness and activation, and perceived productivity, which measures the perceived behavioral output of individuals.

Cronbach's Alpha Table

Scale	Cronbach's Alpha	Interpretation
Productivity (5 items)	0.653	Very Low
Arousal (5 items)	0.714	Acceptable
Reward Anticipation(5 itmes)	0.371	Moderate
Full 15-item Survey	0.712	Acceptable

Fig 1.2 Internal Consistency Reliability Table Summary

Descriptive Statistics

In order to further summarize and provide a strong foundation for the Statistical Analysis, specific descriptive statistics were calculated. Fig 1.3 shows a summary of the descriptive statistics for reward anticipation (RA), psychological arousal (PA), and perceived productivity (PP).

Descriptive Statistics

Statistics	RA	PA	PP
Mean	14.31	24.91	13.72
Median	14.0	25.0	14.0
Modes	15	25	14
Standard Deviation	2.13	6.99	2.82

Fig 2.1 Descriptive Statistics Table Summary

These were calculated with the use of the composite scales; derived from the answers to the 5 questions for each variable. An analysis of the central tendency for each composite scale showed the means and medians were nearly identical, highlighting balanced responses and a fairly symmetrical distribution, as shown by the Kernel Density Estimator (KDE) Curve which shows a smooth continuous representation of the distribution of the composite scales.

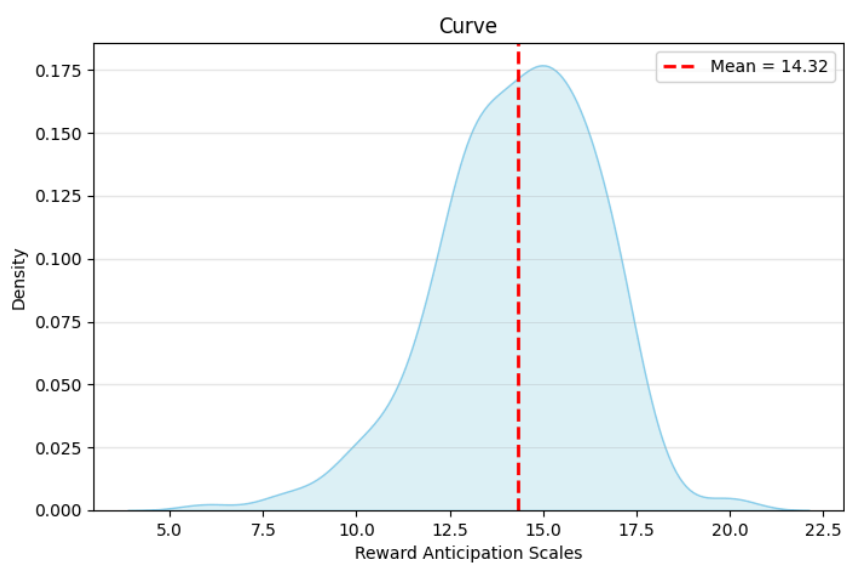


Fig. 2.2 Reward Anticipation Kernel Density Estimator (KDE) Curve

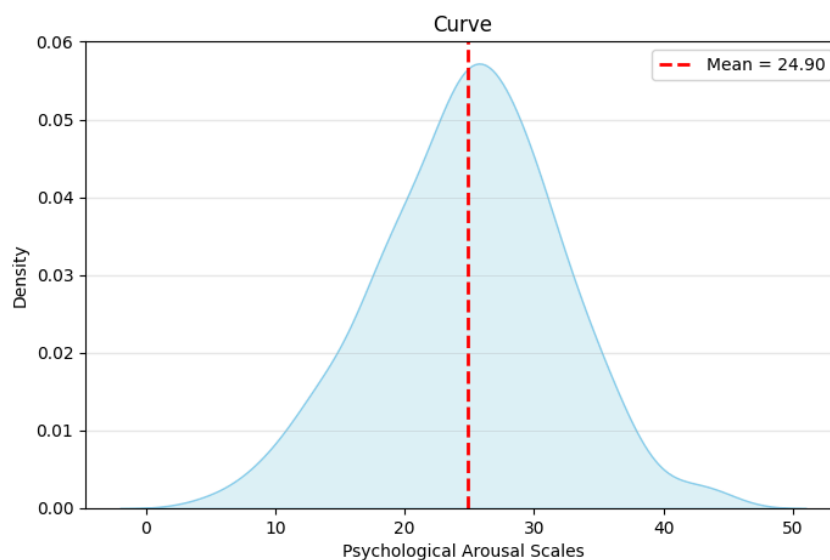


Fig. 2.3 Psychological Arousal Kernel Density Estimator (KDE) Curve

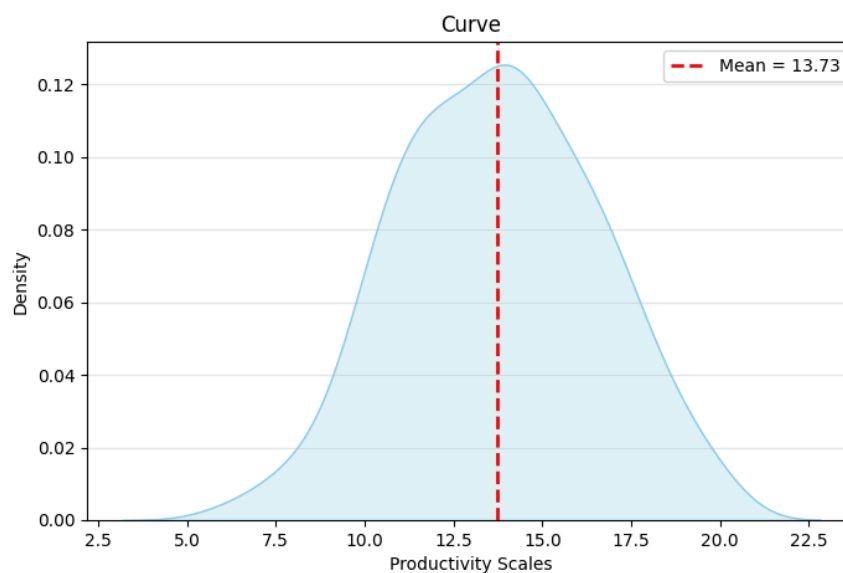


Fig. 2.4 Productivity Kernel Density Estimator (KDE) Curve

Additionally, the Shapiro-Wilk test for normality was conducted to determine whether the three composite scales followed a normal distribution. The null hypothesis: $p \geq 0.05$, was

rejected for both reward Anticipation ($p = 4.31 * 10^{-6}$) and perceived productivity ($p = 0.0084$), which were well below the threshold. Meanwhile for psychological arousal ($p = 0.504$), the null hypothesis was upheld as a normal distribution. While the test resulted in significant statistical deviation from normality for both reward anticipation and perceived productivity, due to the test's sensitivity to larger sample sizes, it does not result in significant skewedness. As a result, it does not rule out the use of a parametric analysis and is stable enough for linear regression and correlation analysis.

In relation to central tendency distribution, about 50% of the responses for reward anticipation and perceived productivity clustered tightly around the median, showing that about half of the participants for reward anticipation clustered around a 3 point difference of each other and a 4 point difference for perceived productivity. Showing that the two constructs were experienced similarly by the participants. Regarding potential outliers, there were two significant outliers at 5 and 8 for reward anticipation and no significant outliers for perceived productivity.

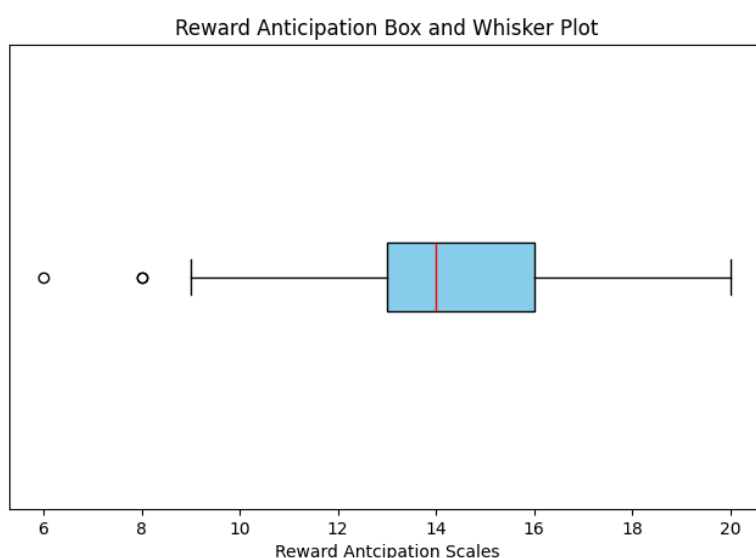


Fig. 2.5 Reward Anticipation Box and Whiskers Plot

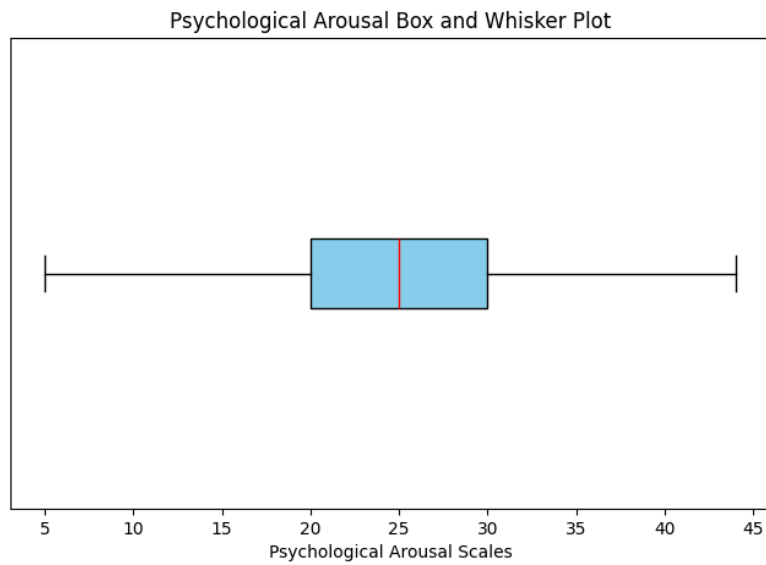


Fig. 2.6 Psychological Arousal Box and Whiskers Plot

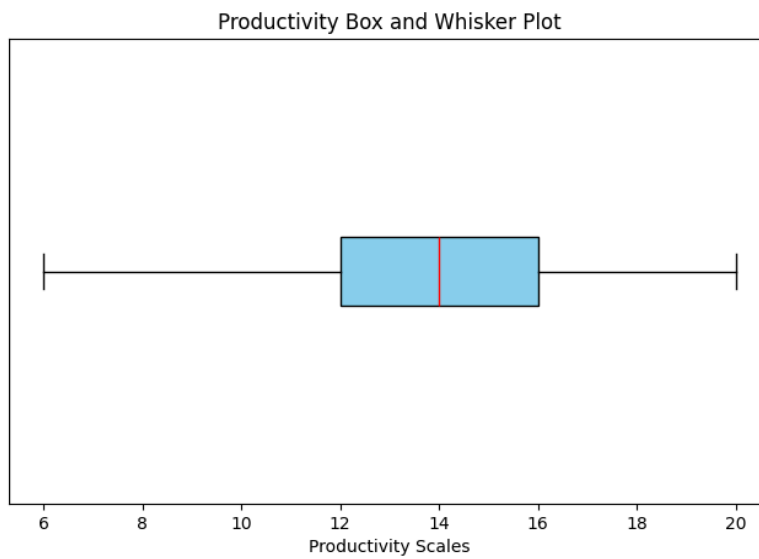


Fig. 2.7 Perceived Productivity Box and Whiskers Plot

For psychological arousal, Fig 2.6, 50% of the responses clustered moderately around the median of 25, showing more variability in the responses compared to reward anticipation. Meaning that individuals tended to experience arousal a bit more differently from each other. There were no significant outliers.

In relation to individual responses, for reward anticipation individuals tended to have an average score of about 14.32 with very little variability of about 2.13. For psychological arousal, the average score was about 24.9 and a moderate variability of about 6.99. While for perceived productivity, individuals tended to score an average of 13.72 with a very little variability of about 2.82. Showing no significant extreme responses to the constructs.

Statistical Analysis

For this research, the Statistical Analysis consisted of the analysis of correlations between the three constructs and independent variables. Specifically for the analysis between the correlations between the three variables; a correlation matrix was constructed.

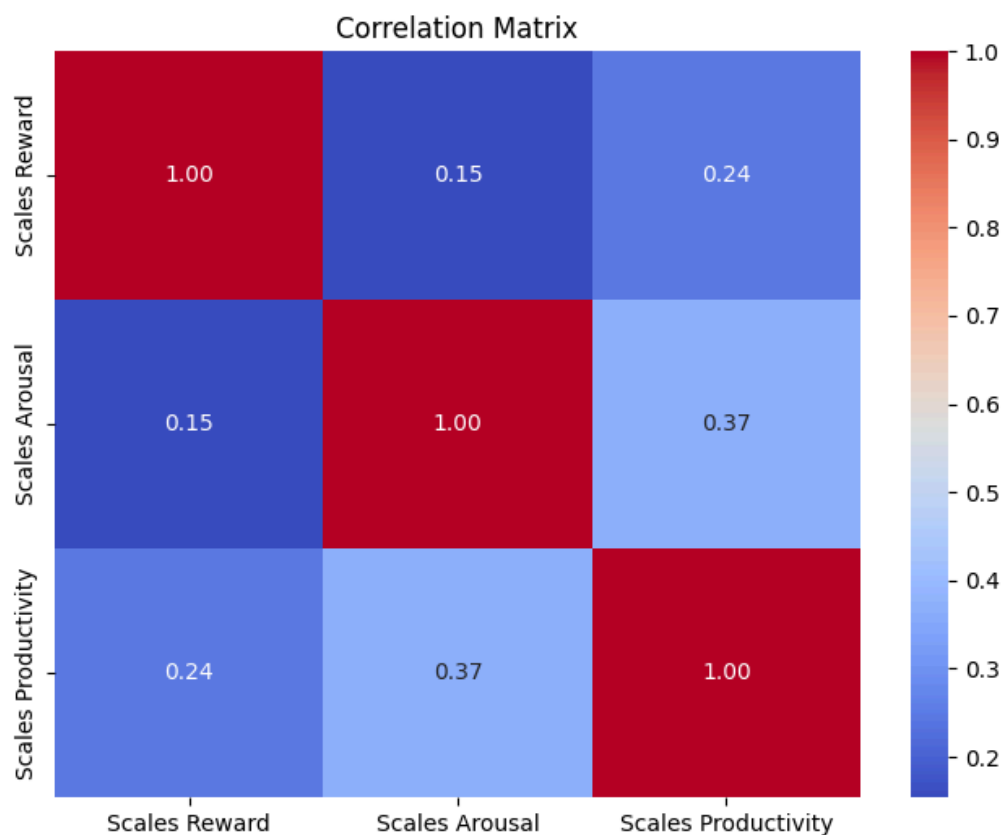


Fig. 3.1 Correlation Matrix

Fig. 3.1 shows a positive association between all three constructs, meaning all constructs move in the same direction. While all associations are significantly low, the strongest association was between Psychological Arousal and Perceived Productivity, with a correlation of about 0.37, highlighting a moderate relationship between them, showing a small but noticeable linear relationship which can predict small changes in perceived productivity. Reward anticipation and perceived productivity showed a small relationship, with a correlation of about 0.24, signifying that any changes in reward anticipation very minimally affect perceived productivity. Meanwhile, Reward Anticipation and Psychological Arousal showed a very weak relationship,

with a correlation of about 0.15, which was expected since these variables are independent of each other.

Gender Relationships

Additionally, specific categories were compared with each other in relation to a specific construct. One of these was the comparison between males and females, which was conducted using a t-test, Fig 3.2.

$$t = \frac{\hat{X}_1 - \hat{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Fig. 3.2 Two Sample - Test

In relation to reward anticipation, there was a significant difference between males and females with a t value of $t = -2.40$, and $p = 0.016$. Females scored higher than males with an average response of about $\hat{X}_2 = 14.669$ compared to males, $\hat{X}_1 = 14.008$, showing that women reacted significantly higher to reward anticipation. Testing the construct of psychological arousal, while females scored higher $\hat{X}_2 = 25.758$ compared to males, $\hat{X}_1 = 24.088$, there was no significant statistical difference between males and females, with $t = -1.834$, and $p = 0.067$. In relation to perceived productivity, females again scored higher than males, $\hat{X}_2 = 14.017$, $\hat{X}_1 = 13.448$, and there was no significant statistical difference between males and females when it comes to productivity, $t = -1.555$, and $p = 0.121$. For psychological arousal and perceived

productivity, there is no meaningful evidence from the data to conclude a difference between males and females.

Academic Relationships

Individual GPA ranges, as collected in the survey, were also compared to each construct through an analysis of variance (ANOVA). These GPA ranges were grouped into great, average, and low ranges, as shown in figure, Fig. 3.4.

GPA Groupings

High	Medium	Low
4.0+		2.5 - 3.0
4.0	3.0 - 3.5	2.0 - 2.5
3.5 - 4.0		Below 2.0

Fig. 3.4 GPA Groupings Table

The analysis revealed that there was a significant academic effect on reward anticipation, with $F = 8.78$, $p = 0.00021$. Individuals with a high GPA scored higher than those with medium and lower GPAs. There was no significant difference between the averages of medium and lower GPAs, highlighting that the main difference was between high GPA individuals and all others. In relation to psychological arousal, there was no evidence of a strong significant difference to conclude that the GPAs differed meaningfully, with $F = 1.10$, $p = 0.334$.

However, there was a very significant difference between high GPA individuals and medium combined with lower individuals, whose averaged responses did not differ significantly.

Highlighting that high GPA individuals responded higher to perceived productivity, almost by 2.6 points, with $F = 26.130$, $p = 7.432 * 10^{-11}$. As a result, it can be stated that there was a positive association between Reward Anticipation and academic performance, and Perceived Productivity and academic performance.

Random Forest

The model was able to explain about 90% of all variance, Best CV $R^2 = 0.901$, and with unseen data, it explains about 87% of all variance, with a Test $R^2 = 0.876$. In relation to the prediction accuracy, Test MAE = 0.623, meaning predictions were off by about 0.623 points, and Test RMSE = 0.799, overall the predictions made by the model differed from the actual observed values by 0.799 points. These metrics show that the model learned real patterns and it is reliable.

The results from the Random Forest model showed that the Perceived Productivity construct with its 5 features dominated, accounting for about 80.66% of the total importance. This shows that Perceived Productivity is mostly self-predicting, or that the best predictor of Perceived Productivity is itself. As a result all other variables contributed significantly less to the total importance, with Psychological Arousal only contributing about 9.7598 %, showing a secondary influence on the predictions. Reward Anticipation contributed significantly less, with a contribution of about 3.7% of the total importance. This highlights that Productivity itself was

the strongest predictor with Psychological Arousal contributing moderately, and Reward Anticipation showing a limited contribution in predicting productivity. This is supported by the statistical analysis shown in Fig. 3.1 which showed that Psychological Arousal and Perceived Productivity were moderately associated, highlighting that individuals who report optimal levels of arousal tend to report being more productive.

Linear Regression

The model was able to explain about 99% of all variance in the data. In relation to prediction accuracy, predictions deviated from real observed values by about 0.014. These metrics showed that the model had a strong internal consistency and its predictions were reliable. The model results showed that perceived productivity was mainly self-prediction accounting for a large weight of about 0.85 . Psychological arousal showed a secondary contribution to predicting productivity. While reward anticipation had an almost negligible contribution to predicting productivity. These results support the findings from the Random Forest model and from the correlation analysis. These showed that psychological arousal had a greater impact on productivity compared to reward anticipation as predictors of productivity. Conveying that individuals who describe being more optimally aroused also reported being more productive. In regards to reward anticipation, the results showed that it had a small almost negligible effect on productivity.

Limitations

The results of this research provided significant results, however there were significant limitations to this research that have to be addressed. Limitations such as a relatively small sample size of 241, which is less than the reasonable sample size of 350. Additionally, the composite scores of each variable were treated to have equal weight, treating every variable as having an equal effect, even if some were stronger than the others. From the results of the Machine Learning models this assumption is correct and as a result limited the accuracy of the findings. Moreover, this study was conducted as a single-point measurement, meaning that responses were taken at a single point in time, this creates a limitation because psychological constructs are volatile and tend to change over time. As a result, individual responses to each construct may be simplified and limited, not accounting for potential meaningful changes in arousal levels or reward feelings throughout the day. Finally, it is also important to note that the survey consisted of self-reported answers, for this reason self-reporting bias and independent confounding variables could have affected the accuracy of the students' responses. These limitations should be considered when determining the significance of the findings.

Future Direction and Conclusion

This study focused on identifying to what extent do reward anticipation and psychological arousal influence productivity in high school students. The findings suggested that internal states as a result of psychological arousal may play a larger role compared to reward anticipation from external rewards in determining productivity. Aligning with the previous findings from the Literature which emphasized the importance of optimal arousal levels to

determine productivity. It also challenged findings that highlighted that reward-based motivation is important in driving motivation to perform well academically.

Future research could expand on these findings and try to understand if the effect of reward anticipation as a predictor of productivity is in fact negligible. Specifically by employing an experimental method to determine the extent to which reward anticipation influences motivation in high school students. Examining the findings from the Literature in relation to the effects of extrinsic and intrinsic motivation. Clarifying the systems in play which affect student performance and build on the results found in this research.

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Appendix A

Reward Anticipation Items

1. When I expect a reward, I feel excited.
2. Thinking about getting strong grades motivates me to study.
3. When I expect a reward, I am better able to complete difficult tasks.
4. Expecting rewards does not improve how well I complete tasks.
5. I perform best when I expect a reward for completing a task.

Arousal Items

1. Based on the images below, how alert or energized do you feel when starting homework?
2. How alert (physical response) or energized do you feel when studying?
3. How alert (physical response) or energized do you feel when you have missing homework?
4. How alert (physical response) or energized do you feel when facing stressful study situations, such as not understanding a topic or being on a close deadline?
5. How alert (physical response) or energized do you feel when expecting to receive a good grade on a test or assignment?

Productivity Items

1. Right now, I feel I am performing at my best in school?
2. I am satisfied with my academic performance in all my classes?
3. Compared to others, I believe that my performance today was above average?
4. When studying, how productive are you?

5.Stress/motivation affect my performance during studying?

Appendix B

Scan to access the Survey



Please answer the following questions honestly based on how you typically feel in the described situations. There are no right or wrong answers.



Appendix C

Information To Relating Survey

- We invite you to participate in this research study. This study examines the relationship between psychological alertness and reward anticipation in relation to academic work, and how these factors are associated with productivity and perceived peak performance. Participation is completely voluntary; it is your choice whether or not you would like to take part in this survey.

- If you decide to participate, you will complete a set of 20 questions, which will take approximately five minutes to complete.
- Participation in the study will take place during the first five minutes of a class period on a school day.
- There are some potential risks associated with participating in this study. These mainly involve topics that may relate to academic stress or emotional discomfort. However, resources will be provided at the end of the survey in case any of these topics cause distress.

INTRODUCTION

You are invited to take part in a research study that explores the relationship between reward anticipation and psychological alertness in relation to academic productivity and perceived performance. This survey is voluntary, and the decision to participate is entirely up to you.

This research evaluates students' perceptions of how alert they feel while studying, how they perform when expecting a reward, and how these factors influence their sense of productivity.

The goal of this study is to better understand how alertness and reward systems may contribute to students' academic performance.

WHAT WILL THE SURVEY CONTAIN

If you decide to participate, you will be asked to complete a short online survey that takes approximately 3 to 5 minutes. You may stop participating at any time by closing the survey.

There is no penalty for choosing not to participate or for stopping before completing the survey.

There are no direct benefits or compensation offered for participating in this study.

RISKS

The potential risks of this study include mild emotional discomfort or stress related to academic topics. Resources related to stress management and academic well-being will be provided at the end of the survey. There may be additional risks that are not currently known.

BENEFITS TO PARTICIPATING IN THIS RESEARCH

While you may not receive any direct personal benefit from participating, this research may help improve understanding of how psychological alertness and reward systems relate to student productivity. This could help inform better study strategies in the future.

CONFIDENTIALITY

Your responses will remain anonymous. The survey does not collect any identifying information, such as names, grades, or personal details. Only the researcher will have access to the survey responses.

If participants would like to see the overall results, a summary of the findings can be shared. No individual participant will be identifiable in any data or reports.

INCENTIVES

There are no incentives or rewards for participating in this survey. Participation is entirely voluntary. You will, however, receive a message of appreciation at the end of the survey thanking you for your time.

YOUR RIGHTS AS A PARTICIPANT

You have the right to not participate in this survey or leave it at any time. Deciding to do so will have no impact or punishment. Feel free to email me if there are any suggestions or concerns, I will gladly assist you in any way possible. The best way to contact me regarding any issues is at student email.